

Sparsity Based Noise Robust Speaker Identification Using a Discriminative Dictionary Learning Approach

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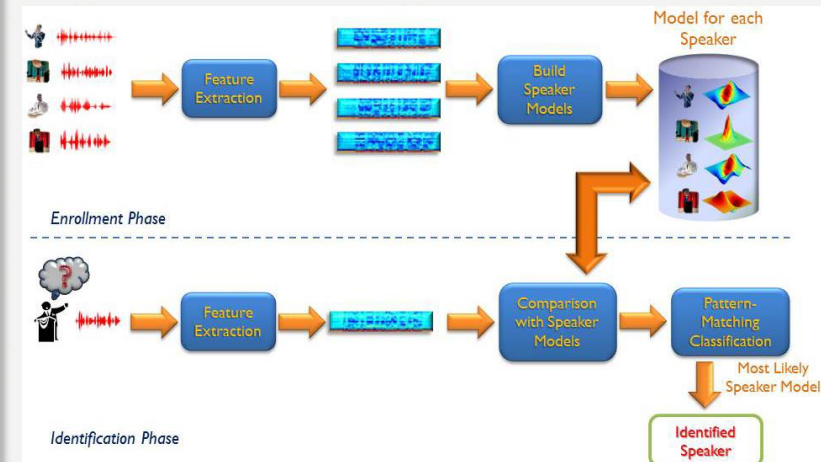


1. Overview

- **Goal** → estimate speaker identity
- Possible applications
 - ✓ information retrieval from speech databases
 - ✓ control of financial transactions
 - ✓ speaker diarization for meetings
 - ✓ speaker-centric ambient intelligence environments



- Focus on current work
 - ✓ text-independent
 - ✓ short training speech data
 - ✓ highly noisy environments



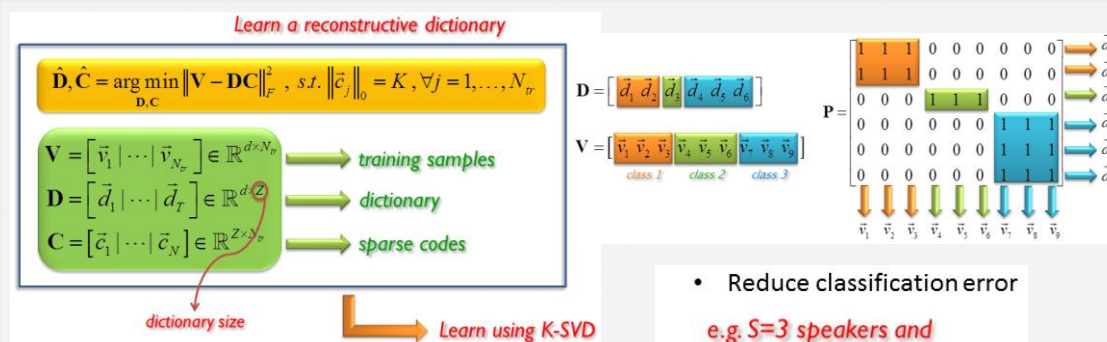
2. ...SRC for speaker identification

- Given $\mathbf{V}$ and $\vec{x}_t$ estimate sparse code $\vec{c}$ by solving the optimization problem (e.g. using OMP)

$$\hat{\vec{c}} = \arg \min_{\vec{c}} \|\vec{x}_t - \mathbf{V}\vec{c}\|_2, \text{ s.t. } \|\vec{c}\|_0 = K$$

$$\left\{ \begin{array}{l} \|\vec{x}_t - \mathbf{V} \cdot \delta_1(\hat{\vec{c}})\|_2 \\ \|\vec{x}_t - \mathbf{V} \cdot \delta_2(\hat{\vec{c}})\|_2 \\ \|\vec{x}_t - \mathbf{V} \cdot \delta_3(\hat{\vec{c}})\|_2 \\ \|\vec{x}_t - \mathbf{V} \cdot \delta_S(\hat{\vec{c}})\|_2 \end{array} \right\} \quad i^* = \arg \min_i \|\vec{x}_t - \mathbf{V} \cdot \delta_i(\hat{\vec{c}})\|_2, i = 1, \dots, S$$

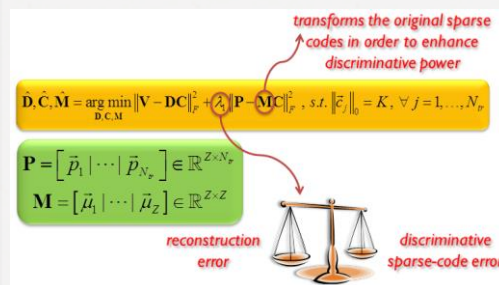
3. Discriminative dictionary sparse coding



- Reduce classification error

e.g. $S=3$ speakers and $N_t=5$ training samples

$$\mathbf{H} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$



$$\hat{\mathbf{D}}, \hat{\mathbf{C}}, \hat{\mathbf{M}}, \hat{\mathbf{B}} = \arg \min_{\mathbf{D}, \mathbf{C}, \mathbf{M}, \mathbf{B}} \left[\|\mathbf{V} - \mathbf{D}\hat{\mathbf{C}}\|_F^2 + \lambda \|\mathbf{P} - \hat{\mathbf{C}}\hat{\mathbf{M}}\|_F^2 + \lambda \|\mathbf{H} - \hat{\mathbf{C}}\hat{\mathbf{B}}\|_F^2, \text{ s.t. } \|\vec{c}_j\|_0 = K, \forall j=1, \dots, N_p \right]$$

- Given test sample $\vec{x}_t$
 - ✓ Compute sparse representation using the estimated dictionary

$$\hat{\vec{c}} = \arg \min_{\vec{c}} \|\vec{x}_t - \hat{\mathbf{D}}\vec{c}\|_2, \text{ s.t. } \|\vec{c}\|_0 = K$$

- ✓ Estimate speaker's identity using the estimated linear classifier

$$\vec{r} = \hat{\mathbf{B}}\hat{\vec{c}} \in \mathbb{R}^{S \times 1}$$

$$i^* = \arg \max_i \vec{r}(i), i = 1, \dots, S$$

2. SRC for speaker identification...

$$\mathbf{V} = \begin{bmatrix} \mathbf{V}_1 & \mathbf{V}_2 & \dots & \mathbf{V}_S \end{bmatrix} = \begin{bmatrix} \vec{v}_{1,1} & \dots & \vec{v}_{1,N_1} & \vec{v}_{2,1} & \dots & \vec{v}_{2,N_2} & \dots & \vec{v}_{S,1} & \dots & \vec{v}_{S,N_S} \end{bmatrix} \in \mathbb{R}^{d \times N_t}$$

belongs to the i^{th} speaker

$$\vec{x}_t = c_{i,1} \vec{v}_{i,1} + c_{i,2} \vec{v}_{i,2} + \dots + c_{i,n_i} \vec{v}_{i,n_i} = \mathbf{V}_i \vec{c}_i$$

4. Experimental evaluation

Dataset

- VOICES corpus → 12 speakers
- Original Fs @ 22kHz →

Feature extraction

- Analysis window → 40 msec
- Overlapping → 50%
- Mel-frequency spectrogram → 40 bands

Training data → 10 sec (multicondition training)

Testing data → 15.6 sec

- add noise @ 0, 5, 10, 15, 20 dB
- white, speech babble, car engine, factory floor (NOISEX92)

